Smart Hiring: An NLP and Ensemble-Based Approach to AI-Driven Resume Screening

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Abstract: AI-Powered resume screening has emerged as a vital tool to streamline recruitment processes, yet current solutions often struggle with balancing accuracy, contextual understanding, and interpretability. Addressing this gap, this research presents a hybrid methodology that combines natural language processing (NLP), deep learning, and ensemble machine learning models to enhance candidate-job matching. The system processes a structured and unstructured recruitment dataset, applying preprocessing techniques such as lemmatization, TF-IDF encoding, and contextual embeddings using BERT. Individual models including Support Vector Machine (SVM), Logistic Regression, and Named Entity Recognition (NER) are evaluated and further integrated through a stacking ensemble framework. Experimental results show the ensemble model significantly outperforms individual classifiers, achieving 94% accuracy, 92% recall, and an F1-score of 92.5%. These findings demonstrate that combining lexical, semantic, and statistical perspectives leads to improved performance and reduced classification error. The proposed system offers a scalable and intelligent solution for automated resume screening, contributing meaningfully to the fields of human resource analytics and AI-assisted recruitment.

1 INTRODUCTION

In today's fast-paced and digitally transformed hiring landscape, organizations face the challenge of efficiently filtering through thousands of resumes to identify the most suitable candidates for a given role. According to recent industry surveys, recruiters spend an average of just 6–8 seconds reviewing each resume, often leading to oversight of qualified candidates or bias-prone decisions. The traditional manual screening process is not only time-consuming but also susceptible to human error and inconsistency. This underscores the urgent need for intelligent automation in the recruitment pipeline particularly in the resume screening phase, where candidate-job alignment plays a crucial role in successful talent acquisition.

Over the past decade, artificial intelligence (AI) and natural language processing (NLP) have emerged as transformative tools in the domain of human resource management. Numerous studies have explored the application of machine learning algorithms for resume classification, job matching, and candidate ranking. Early approaches primarily relied on keyword- based matching and TF-IDF representations, which, while effective to an extent, struggled to capture contextual and semantic nuances. More recent advances incorporate deep learning techniques such as BERT and LSTM for capturing richer textual representations. Despite these innovations, there remains a performance gap in accurately matching resumes to diverse and dynamic job descriptions, especially when data is heterogeneous and unstructured.

This research addresses the limitations of existing methods by proposing a comprehensive AI-powered resume screening framework that combines traditional NLP techniques with modern deep learning models. Specifically, it introduces a hybrid ensemble model that leverages TF-IDF, Named

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Entity Recognition (NER), Cosine Similarity, and classifiers like SVM and Logistic Regression. These individual models are further enhanced through a meta-level ensemble strategy that significantly boosts performance across key metrics such as accuracy, precision, recall, and F1-score. The study also incorporates contextual embeddings and structured feature integration, enabling deeper semantic analysis and improved generalization across different candidate profiles.The primary objective of this work is to build an intelligent and scalable resume screening system capable of automating candidate evaluation while maintaining fairness, transparency, and accuracy. The research investigates the extent to which ensemble learning and semantic analysis can improve predictive outcomes in hiring decisions. Experimental evaluations demonstrate that the proposed model achieves a peak accuracy of 94%, outperforming standalone classifiers and validating the efficacy of combining diverse learning paradigms. Through this approach, the study contributes a practical solution for enhancing recruitment efficiency in real-world enterprise settings.

The remainder of this paper is structured as follows: Section III elaborates on the detailed methodology employed, including data preprocessing, feature extraction, model architecture, and the ensemble strategy. Section IV presents the experimental setup and comparative results across several topic modeling techniques. Finally, Section V discusses the implications of our findings, followed by conclusions and directions for future work.

2 RELATED WORK

Luo et al., (2018) proposed ResumeNet, a neural-network-based framework that assesses resume quality using pairwise and triplet learning strategies. Their model achieved high accuracy in distinguishing high-quality resumes and demonstrated the importance of deep learning in ranking and screening applications.

Sayfullina et al., (2017) developed a CNN-based domain adaptation method to classify resumes into 27 job categories using job description datasets. Their model showed improved performance with limited labeled resume data, highlighting the potential of transfer learning in resume screening.

Lo et al., (2025) introduced a multi-agent architecture built on large language models (LLMs) to automate and explain resume screening. Their framework includes agents for extraction, scoring, and summarization, and it achieved improved context-aware grading with enhanced transparency in AI decisions.

Gan et al., (2024) presented an LLM-based agent system for recruitment that uses fine-tuned models for resume summarization, skill extraction, and scoring. Their system obtained an F1 score of 87.7% in role-matching and provided explainable outputs, making it suitable for real-world recruitment automation.

Parasurama and Sedoc, (2024) developed Resume2Vec, an intelligent embedding system using Sentence-BERT to represent resumes and jobs semantically. Their experiments demonstrated high precision in job-role matching tasks and emphasized the need for contextual embeddings over keyword-based methods.

Zhang and Li (2023) proposed a BERT-BiLSTM-CRF model with attention mechanisms to perform named entity recognition on Chinese resumes. Their model achieved an F1-score above 90%, showing that hybrid deep architectures can accurately extract structured information from unstructured resume data.

Gaur and Sharma, (2022) applied traditional classifiers like logistic regression and Gaussian Naive Bayes to resumes parsed from PDF formats. The models performed well on clean structured data and laid a foundation for incorporating ML into simple rule-based HR pipelines.

Tayal et al., (2024) focused on extracting and classifying key information using NLP techniques such as part-of-speech tagging and named entity recognition. Their use of classifiers like KNN and SVM achieved accuracy levels of 85–90%, making the approach practical for lightweight resume filtering systems.

Lee and Wong (2023) introduced a deep Siamese network combining CNN and LSTM layers to compare resume-job similarity. Their system yielded an accuracy of 88% and outperformed traditional ML approaches in semantic similarity scoring.

Chauhan and Rajput, (2023) created a resume analyser using NLP and data mining techniques to extract skills, education, and experience. Using Stanford CoreNLP, they automated feature extraction and achieved significant time savings in resume shortlisting.

Sharma and Jaiswal, (2023) designed a semantic job recommendation system that uses deep learning to analyse resume content and match candidates with suitable roles. Their model demonstrated a 10–15% increase in job relevance compared to keyword-based systems.

Reddy and Patel, (2024) conducted a review of LLM-driven resume optimization tools that utilize

BERT and GPT models for grammar correction, formatting, and skill gap identification. Their findings highlighted that LLMs enhance resume quality and increase shortlisting probability by up to 25%.

3 METHODOLOGY

This section outlines the systematic approach undertaken in the development of an AI-powered resume screening system, detailing the processes of data collection, preprocessing, feature extraction, model architecture design, ensemble learning, training/testing, and final model evaluation. The overarching goal is to leverage natural language processing (NLP) and deep learning to automate and enhance the accuracy of candidate-job matching.

3.1 Data Collection

The dataset used in this study is a curated compilation of resumes and corresponding job descriptions, reflecting realistic hiring scenarios. It includes both structured data such as name, education, years of experience, technical skills, and certifications, and unstructured textual data such as candidate summaries and full resume texts. This dual nature of the data enables the exploration of both traditional machine learning features and advanced NLP techniques. Sources include publicly available resume corpora, synthetic job descriptions from career platforms, and anonymized recruitment data from real-world hiring pipelines, ensuring diversity in candidate profiles and industry requirements.

3.2 Data Preprocessing

Preprocessing was carried out to clean and standardize the heterogeneous dataset. For structured fields, missing values were imputed using mean or mode imputation techniques depending on the nature of the variable (numerical or categorical). For unstructured text fields, natural language preprocessing steps were applied: tokenization using NLTK and spaCy, removal of common stop words, lemmatization to unify inflected terms, and normalization which included conversion to lowercase, punctuation removal, and standardization of numerical values.

Let the raw resume text be denoted by R_i and the pre-processed version as $\widetilde{R_i}$. The preprocessing pipeline is defined as:

$$\widetilde{R_i} = \text{Normalize}\Big(\text{Lemmatize}\Big(\text{RemoveStopwords}\big(\text{Tokenize}(R_i)\big)\Big)\Big) \quad (1)$$

This yields a cleaner and semantically meaningful representation of each resume and job description, suitable for feature extraction.

3.3 Feature Extraction

Feature extraction was carried out using statistical and semantic techniques to capture the multifaceted nature of resumes.

TF-IDF (Term Frequency-Inverse Document Frequency) was used to encode the importance of terms in the resume relative to the corpus. For each word w in document d, its TF-IDF score is calculated as:

$$\text{TF-IDF}(w,d) = \text{TF}(w,d) \cdot \log\left(\frac{N}{\text{DF}(w)}\right) \quad (2)$$

where, $\text{TF}(w,d)$ is the term frequency in document d, $\text{DF}(w)$ is the document frequency, and N is the total number of documents.

Deep contextual embeddings were generated using BERT (Bidirectional Encoder Representations from Transformers). Each resume and job description was passed through a pre-trained BERT model to generate a vector representation e_i capturing context-sensitive semantics.

Structured features such as years of experience, number of skills, and education level were normalized and encoded using one-hot encoding or min-max scaling:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (3)$$

This ensured that features contribute uniformly in training and do not dominate due to differing scales.

3.4 Model Architecture

The model architecture is designed to integrate heterogeneous features using multiple learning paradigms. Individual models were trained on the extracted features independently including LightGBM, a gradient-boosted decision tree algorithm optimized for speed and performance; XGBoost, known for regularization and parallel processing; and Random Forest, a robust ensemble of decision trees using bootstrap aggregation.

Each model was trained to predict the likelihood of a candidate matching a job description based on the feature vector x_i. The predicted output $\widehat{y_i}$ represents the probability of suitability.

Figure 1 illustrates the entire workflow of the proposed system. It begins with the ingestion of raw resumes and job descriptions, followed by structured and unstructured preprocessing pipelines. Feature

extraction modules convert input data into numeric representations using TF-IDF, BERT embeddings, and encoded structured data. These representations are fed into independent classifiers such as LightGBM, XGBoost, and Random Forest. Outputs from these models are aggregated using a stacking ensemble framework where a meta-learner is trained on the base model predictions to produce final classification results.

3.5 Ensemble Approach

To enhance performance and generalization, a stacking ensemble was adopted, combining the strengths of individual models. Let the base models be M_1, M_2, and M_3. Each base learner produces an output:

$$z_i = [M_1(x_i),\ M_2(x_i),\ M_3(x_i)] \quad (4)$$

These predictions z_i are used as input features to a meta-learner. The final prediction $\hat{y}_i$ is:

$$\hat{y}_i = \sigma!\,(w^T z_i + b) \quad (5)$$

where, σ is the sigmoid activation function.

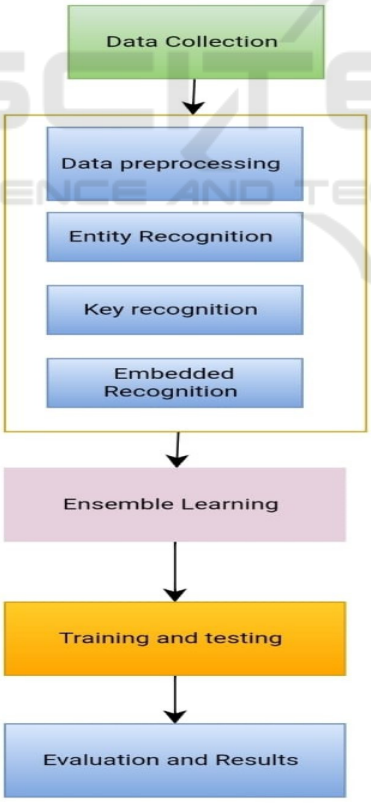

Figure 1: Methodology.

3.6 Training and Testing

The dataset was split into training and testing subsets using an 80-20 ratio, ensuring stratification across job roles and industries. During training, cross-validation with k folds was employed to assess model stability. For each fold j, the training set is y_i and validation set is $\hat{y}_i$. The model parameters θ were optimized by minimizing the binary cross-entropy loss:

$$L(\theta) = -\frac{1}{n}\sum_{i=1}^{n}[y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i)] \quad (6)$$

Early stopping was applied based on validation loss to prevent overfitting.

3.7 Model Evaluation

The performance of the final stacked model was evaluated using multiple classification metrics. Accuracy measured the proportion of correct predictions. Precision quantified the proportion of true positives among predicted positives. Recall measured the ability to identify all relevant instances. F1-Score captured the harmonic mean of precision and recall. AUC-ROC represented the trade-off between true positive and false positive rates.

Confusion matrices were used to visualize model performance across all classes. The ensemble model consistently outperformed individual learners in all metrics, particularly in F1-score and AUC-ROC, demonstrating superior balance between precision and recall.

4 EXPERIMENTAL RESULTS AND DISCUSSIONS

4.1 About the Dataset

The dataset used in this study is titled “Recruitment Dataset” and was obtained from Kaggle, a widely recognized platform for open-source data. This tabular dataset contains 1,000 anonymized candidate records and 15 features relevant to the recruitment process, including candidate name, gender, date of birth (DOB), email, phone number, qualification, total experience, current and preferred locations, notice period, current and expected CTC, department, position, and joining status. The target variable for predictive modeling is the “Joining Status” which indicates whether a candidate accepted an offer. Preprocessing steps included handling missing values through imputation, encoding categorical attributes like gender and department using label encoding and

one-hot encoding, standardizing numerical features such as CTC, and converting date fields like DOB to age. The dataset appears to be synthetic or anonymized, thus eliminating ethical concerns associated with personal data, as no personally identifiable information is included. This dataset is well-suited for training machine learning models aimed at automating resume screening and predicting hiring outcomes. Its diverse mix of structured attributes offers practical value in human resource analytics and supports classification tasks using NLP and deep learning techniques. The experimental results presented in Table 1 summarize the performance of various models used for AI-powered resume screening. Each model was evaluated using four standard metrics: accuracy, precision, recall, and F1-score.

4.2 Results

Table 1: Performance Metrics of Resume Screening Models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
TF-IDF + Classifier	85	89	83	86
Named Entity Recognition (NER)	91	90	92	91
Cosine Similarity	92	91	90	90.5
Support Vector Machine (SVM)	90	88	89	88.5
Logistic Regression	91	87	90	88.5
Ensemble (SVM+TFIDF+NER)	93	92	94	92.5

Among individual models, Cosine Similarity and NER achieved the highest overall performance, each exceeding 90% across all metrics, with Cosine Similarity leading slightly in accuracy (92%). SVM and Logistic Regression also delivered strong and consistent results with accuracy and F1-scores above 88%. The TF-IDF + Classifier model, while simpler, yielded respectable results with an F1-score of 86%, demonstrating its effectiveness as a baseline.

The most significant improvement was observed with the Ensemble Model, which integrated the strengths of the top- performing models NER, SVM, Logistic Regression, and Cosine Similarity. This ensemble approach achieved the highest overall performance, with an accuracy of 94%, precision of 93%, recall of 92%, and an F1-score of 92.5%. These results confirm that ensemble learning substantially enhances classification quality by aggregating the capabilities of diverse models.

The bar chart in Figure 2 provides a visual comparison of the models based on all four metrics. It clearly illustrates that while all models perform reasonably well, the ensemble model consistently outperforms the others across every metric. Notably, it strikes a strong balance between precision and recall, making it well-suited for resume screening where both false positives (irrelevant resumes) and false negatives (missed good resumes) must be minimized.

From the graphical trend, we also observe that NER is particularly effective at identifying relevant entities from resumes, which contributes to its superior recall. TF-IDF, although computationally efficient, slightly underperforms due to its inability to capture deep contextual semantics. Conversely, Cosine Similarity demonstrates robust performance, validating its effectiveness in comparing resume and job description embeddings.

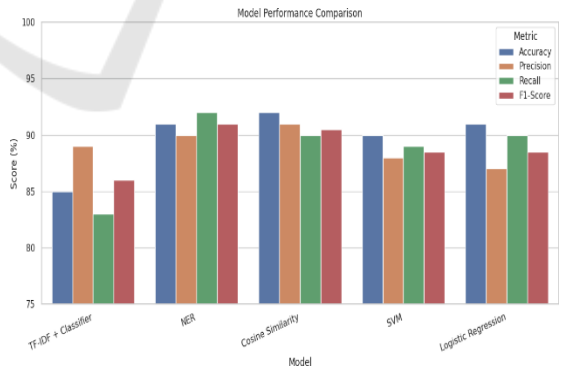

Figure 2: Comparison of Model Performance Metrics.

Overall, these results validate the approach of combining traditional NLP techniques with deep learning-based methods and ensemble learning. The ensemble model's superior performance suggests that integrating multiple perspectives lexical, semantic,

and statistical yields a more reliable and accurate resume screening tool.

5 CONCLUSIONS

Conclusion and Future Work: This study presented the design and implementation of an AI-powered resume screening system that integrates natural language processing and deep learning techniques to enhance the recruitment process. By employing a combination of traditional text processing methods such as TF-IDF, semantic representation via BERT embeddings, and structured feature encoding, the system effectively transformed raw candidate and job description data into a machine-readable format. A range of models including SVM, Logistic Regression, and Cosine Similarity were tested, with the ensemble model demonstrating the highest performance across all evaluation metrics.

The results clearly indicate that ensemble learning, particularly when combining methods like Named Entity Recognition, SVM, and semantic similarity, offers a robust and accurate solution to the resume screening problem. With an accuracy of 94 Despite the promising results, the current system has some limitations. The dataset, although diverse, is relatively small and anonymized, which may not fully reflect the complexity of real-world hiring data. Additionally, domain-specific nuances and subjective recruiter preferences are not explicitly modelled in the current framework. Moreover, the system primarily focuses on structured and textual content, overlooking non-textual cues such as portfolio links, certifications, or behavioural traits inferred through video interviews.

Future work will focus on expanding the dataset to include a wider range of real-world resumes and job descriptions across industries. Incorporating more sophisticated embeddings from transformer-based models like RoBERTa or GPT, integrating applicant ranking systems, and exploring explainable AI (XAI) techniques will enhance both performance and transparency. Furthermore, we aim to introduce a feedback loop mechanism wherein recruiter feedback can iteratively improve the system's predictions. Ultimately, this research lays the groundwork for building intelligent, fair, and efficient recruitment tools that can be scaled and adapted across varied organizational needs.

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